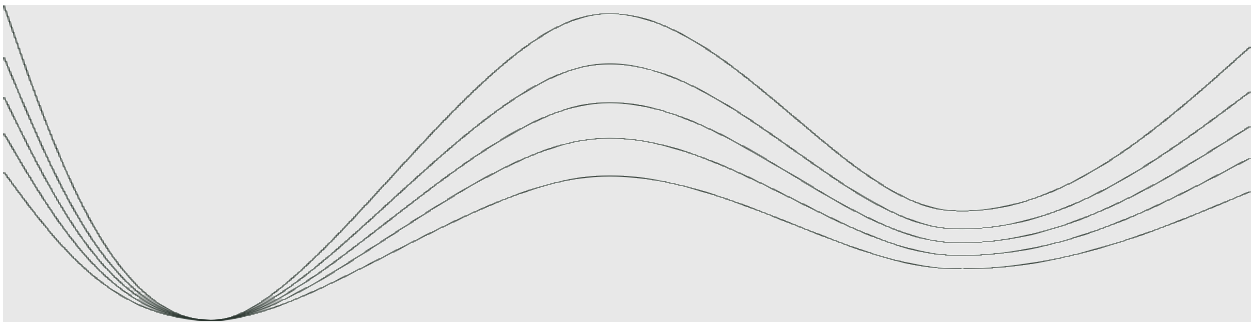

Prophesy for CIB

Manual

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Introduction

Cross-Impact Balances (CIB) analysis was designed for systematically generating scenarios and assessing them for internal consistency. Through succession analysis, CIB also "offer[s] an idea of the direction in which the system might move from its momentary state" (Weimer-Jehle 2006). In support of this goal, Prophesy introduces several improvements to succession analysis related to ties, succession rules, and auxiliary descriptors.

Two or more variants of a descriptor can be tied for the highest balance. The traditional approach to this problem has been to select the variant with the lowest index, which biases the successions. Prophesy gives the user control over how ties are broken. The legacy tiebreaker handles ties in the way described above. An advantage of it is that it ensures deterministic successions. The random tiebreaker selects one of the tied variants at random. The conservative tiebreaker handles two cases: (i) when the current variant is one of the tied variants, the current variant is selected; (ii) otherwise, the random tiebreaker is used.

The conservative and random tiebreakers introduce stochasticity into succession analysis. Since the outcome of succession analysis is no longer deterministic, Prophesy allows the user to run multiple succession trials per initial scenario. The modal outcome is then taken as the outcome for the tested scenario, be it a point attractor or a cyclical attractor. Stochasticity also complicates the identification of cyclical attractors, since they can be noisy. Prophesy includes strategies for identifying cyclical attractors in noisy successions.

Succession rules govern how descriptors are corrected at each step in a succession analysis. The four most widely used rules are global (simultaneously update each descriptor to its variant with the highest balance), local (update only the descriptor with the greatest inconsistency), incremental (simultaneously update all descriptors, but restrict their movement to an adjacent variant), and adiabatic (cycle through the descriptors in order of their indices, updating only one descriptor at a time) (Weimer-Jehle 2026). Stochastic rules have also been developed (Schweizer et al. 2023). While these rules are suitable for identifying attractors, they are less ideal for modelling system dynamics. Real systems can have multiple processes running in parallel, which neither the local nor the adiabatic rule can represent. Some system elements may undergo radical changes, while others may change incrementally; the incremental rule cannot represent both behaviours. Prophesy implements the global succession rule and devolves decision-making about descriptor behaviour to the descriptors themselves. It does this by introducing descriptor

types. Nominal descriptors can move directly from the current variant to the variant with the highest balance. Ordinal descriptors can move only one variant per succession step in the direction of the variant with the highest balance. This combination of global succession rule and descriptor types allows for finer-grained representations of the dynamics of real systems. In a subsequent release, rates of change will also be incorporated into Prophesy through an Intervals Matrix (currently under development), replacing the need for an adiabatic succession rule.

Auxiliary descriptors are used to represent impacts that depend on the variants of two or more other descriptors (Weimer-Jehle 2009). For example, in the Easterlin paradox, happiness increases with GDP growth when GDP levels are low, but this effect weakens or disappears at higher GDP levels (Easterlin 1974). Neither GDP nor GDP growth alone can adequately represent this relationship. Happiness increases when GDP is both low and growing, meaning that the effect depends on the combination of the two descriptors. To represent such combined effects, an auxiliary descriptor is introduced whose variants correspond to the Cartesian product of the variants of the associated descriptors. Each variant of the auxiliary descriptor therefore represents a specific combination of the variants of the associated descriptors. Traditionally, the variant of an auxiliary descriptor is controlled by specifying cross-impacts on the auxiliary descriptor so that it is corrected at the next succession step (Weimer-Jehle 2009). However, this approach introduces delays and can distort successions, because the influence of the auxiliary descriptor should occur in the same step as the changes in its associated descriptors. Prophesy avoids this problem by updating the variant of the auxiliary descriptor programmatically within the same step, rather than relying on cross-impacts and succession analysis. A simple check is performed: in the current step, is the variant of the auxiliary descriptor consistent with the variants of its associated descriptors? If not, the auxiliary descriptor is updated accordingly. Proper handling of auxiliary variables extends CIB beyond pairwise causal relationships (e.g., A → B) to more complex causal forms involving Boolean operators (e.g., A and B → C), which is necessary for contextualizing causal relationships (e.g., A → B holds if and only if C is present).

Through improved handling of ties, succession rules, and auxiliary descriptors, Prophesy makes CIB successions more realistic representations of the dynamics of real systems. Prophesy includes a variety of other features:

- **Perturbation analysis:** For a degree of perturbation (number of variants replaced simultaneously), Prophesy can generate all neighbours of a set of initial scenarios and subject them to succession analysis. The results are then presented in a Transformation Matrix with the initial scenarios in the first column and the attractors

in the first row. The remaining cells of the matrix contain combinations of variant replacements that cause the initial scenario to evolve into the attractor. Perturbation analysis was developed by Kearney (2021) and is similar to the simulated annealing approach developed by Kemp-Benedict et al. (2019).

- **Pathway analysis:** For a given pair of initial scenario and attractor from the results of a perturbation analysis, Prophesy can report all pathways, which are subsequences of succession trials from the initial scenario to the attractor. Pathways are reported with summary statistics and can also be scored (e.g., desirability).
- **Sensitivity analysis:** Prophesy can generate multiple versions of a base model. Users can specify their confidence in the cross-impacts by judgment section, affecting the proportion of model versions in which those cross-impacts are included. Users can also specify the proportion of model versions in which the cross-impacts of judgment sections are deleted, have their signs flipped, or are multiplied by a specified multiplier.
- **Discrete Maximin designs:** Depending on the settings used, sensitivity analysis can involve sampling from a large space of possible model versions. To ensure diverse coverage of this space, Prophesy uses a discrete maximin design. This approach is also used for sampling from the scenario space of very large models.
- **Descriptive statistics and diagnostics:** Prophesy provides several new descriptive statistics and diagnostics for scenario analysis, perturbation analysis, and pathway analysis, such as minimum and average Hamming distance, trial convergence, model convergence, trial coverage, and trial stability.

Prophesy is under active development (see Part 4 of this manual). Please report any bugs or feature requests to norman.kearney@unibe.ch.

Part 1: CIB Fundamentals

1.1 Descriptors, variants, and cross-impacts

CIB models consist of descriptors, variants, and cross-impacts. **Descriptors** are categorical variables representing system elements and their possible, mutually exclusive states, called **variants**. Variants interact with one another via numerical **cross-impacts**, where the impact of variant x_i of descriptor X on variant y_j of descriptor Y expresses how much the *plausibility* of y_j changes given x_i relative to the other variants of Y.

CIB models typically consist of multiple descriptors, with each descriptor affecting and being affected by one or more other descriptors. A descriptor is a **target descriptor** when it receives an impact, and a **source descriptor** when it gives an impact. Likewise, a variant is a **target variant** when it receives an impact, and a **source variant** when it gives an impact.

1.2 Matrices, judgment sections, judgment groups, and impact columns

A CIB model is represented as a symmetrical hypermatrix, with the descriptors and variants running down the lefthand columns and the top rows. CIB matrices consist of judgment sections, judgment groups, and impact columns. A **judgment section** is a sub-matrix consisting of all cells at the intersection of a source descriptor and a target descriptor. A **judgment group** is a row within a judgment section. An **impact column** is the set of all cells in the column of a target variant.

Impact column (cross-impacts = relative strength of source variants)

		X_1				X_2				Y		
		x_{11}	x_{12}	x_{13}		x_{21}	x_{22}	x_{23}		y_1	y_2	y_3
X_1	x_{11}	0	0	0		0	0	0		2	1	-3
	x_{12}	0	0	0		0	0	0		0	0	0
	x_{13}	0	0	0		0	0	0		0	0	0
X_2	x_{21}	0	0	0		0	0	0		-2	0	2
	x_{22}	0	0	0		0	0	0		0	0	0
	x_{23}	0	0	0		0	0	0		0	0	0
Y	y_1	1	2	-3		0	0	0		0	0	0
	y_2	0	0	0		1	0	-1		0	0	0
	y_3	0	0	0		0	0	0		0	0	0

Judgment section

Judgment group (cross-impacts = relative plausibility of target variants)

1.3 Relative plausibility of cross-impacts

Cross-impacts are zero-centred: positive impacts represent increases in plausibility and negative impacts represent decreases. Because plausibility is purely relative within a set of mutually exclusive variants, increases in plausibility must be offset by decreases elsewhere. A cross-impact of zero means "neither more nor less plausible", so positive impacts are meaningless unless compensated by negative ones and vice versa. To ensure internal coherence and comparability, cross-impacts within a judgment group (i.e., the impacts of one source variant on all variants of a target descriptor) must sum to zero.[1] Thus, the cross-impacts within a judgement group give the **relative plausibility** of the variants and comprise a **plausibility distribution** for the judgment group.

To see why, suppose we have two descriptors, X and Y, each with three mutually exclusive variants [x_1,x_2,x_3] and [y_1,y_2,y_3]. Suppose that given x_1, variants y_1 and y_2 are plausible, with y_1 more plausible than y_2.

In Example 1, the cross-impact of x_1 on y_1 is 2, on y_2 is 1, and on y_3 is 0. This configuration is incoherent, as the increase in plausibility of y_1 and y_2 is not offset by a decrease in plausibility of y_3.

In Example 2, this coherence is restored: if x_1 increases the plausibility of y_1 by 2 and y_2 by 1, it must decrease the plausibility of y_3 by -3.

In Example 3, the cross-impact of x_1 on y_3 is -2. This leaves a residual +1 increase in plausibility that is not accounted for. The increase in plausibility of y_1 and the decrease in y_3 cancel out, but the increased plausibility of y_2 remains undefined – increased relative to which alternative?

Only Example 2 is coherent: given x_1, y_1 is most plausible, y_2 less so, and y_3 implausible.

		Y		
		y_1	y_2	y_3
X	x_1	2	1	0
	x_2	0	0	0
	x_3	0	0	0

Example 1: Incoherent cross-impacts

		Y		
		y_1	y_2	y_3
X	x_1	2	1	-3
	x_2	0	0	0
	x_3	0	0	0

Example 2: Coherent cross-impacts

		Y		
		y_1	y_2	y_3
X	x_1	2	1	-2
	x_2	0	0	0
	x_3	0	0	0

Example 3: Incoherent cross-impacts

[1] See exception below.

1.4 Scenarios, balances, and consistency

The overall state of a CIB model is represented by a **scenario**, which consists of exactly one variant per descriptor. Depending on the variants that they contain, scenarios may be more or less internally consistent.

Internal consistency is a measure of the degree to which the variants of a scenario are mutually reinforcing and serves as an indicator of the plausibility of the scenario. Internal consistency is determined by calculating the balances of the variants and the consistencies of the descriptors, and then by summing the consistencies across all descriptors.

The **balance** of a variant is calculated by summing the cross-impacts applied to the variant by the other variants in the scenario. In Example 4, there are two descriptors affecting Y: X_1 and X_2. Consider the scenario $\{x_{11}, x_{21}, y_1\}$. The cross-impact of x_{11} on y_1 is 2, while the cross-impact of x_{21} on y_1 is -2. Thus, the balance for y_1 is 2 + (-2) = 0.

The **consistency** of a descriptor is calculated by subtracting the maximum balance among its variants from the balance of the variant present in the scenario. This measures how far the current variant deviates from the most supported alternative. In Example 4, the variant of Y with the highest balance is y_2 with a balance of 1, while the balance of the current variant y_1 is 0. The consistency of Y in the scenario $\{x_{11}, x_{21}, y_1\}$ is 0 – 1 = -1.

		X_1				X_2				Y		
		x_{11}	x_{12}	x_{13}		x_{21}	x_{22}	x_{23}		y_1	y_2	y_3
X_1	x_{11}	0	0	0		0	0	0		2	1	-3
	x_{12}	0	0	0		0	0	0		0	0	0
	x_{13}	0	0	0		0	0	0		0	0	0
X_2	x_{21}	0	0	0		0	0	0		-2	0	2
	x_{22}	0	0	0		0	0	0		0	0	0
	x_{23}	0	0	0		0	0	0		0	0	0
Y	y_1	1	2	-3		0	0	0		0	0	0
	y_2	0	0	0		1	0	-1		0	0	0
	y_3	0	0	0		0	0	0		0	0	0
Balances		1	2	-3		0	0	0		0	1	-1

Example 4: Balances and consistencies

By calculating the balance of each variant and the consistency of each descriptor, we obtain the **internal consistency** of the scenario $\{x_{11}, x_{21}, y_1\}$. The consistency of X_1 is 1 - 2 = -1, and the consistency of X_2 is 0. Thus, the internal consistency of the scenario $\{x_{11}, x_{21}, y_1\}$ is -1 + -1 = -2.

1.5 Successors and succession analysis

Inconsistencies are used to determine the **successor** (or successors) of a scenario, that is, the scenario(s) obtained by correcting inconsistencies in the current scenario. A descriptor is inconsistent when the balance of its current variant is lower than the maximum balance among its variants. Inconsistencies are corrected by replacing the variant of each inconsistent descriptor with a variant that has the highest balance.

Example 5 shows the successor of the scenario in Example 4: X_1 shifts from x_{11} to x_{12} and Y shifts from y_1 to y_2. The resulting scenario, $\{x_{12}, x_{21}, y_2\}$, contains new inconsistencies. Example 6 shows the result of resolving these inconsistencies: a stable and internally consistent scenario in which no descriptor is inconsistent. The process of stepping through and correcting inconsistent scenarios is called **succession analysis**.

		X_1				X_2				Y		
		x_{11}	x_{12}	x_{13}		x_{21}	x_{22}	x_{23}		y_1	y_2	y_3
X_1	x_{11}	0	0	0		0	0	0		2	1	-3
	x_{12}	0	0	0		0	0	0		0	0	0
	x_{13}	0	0	0		0	0	0		0	0	0
X_2	x_{21}	0	0	0		0	0	0		-2	0	2
	x_{22}	0	0	0		0	0	0		0	0	0
	x_{23}	0	0	0		0	0	0		0	0	0
Y	y_1	1	2	-3		0	0	0		0	0	0
	y_2	0	0	0		1	0	-1		0	0	0
	y_3	0	0	0		0	0	0		0	0	0
Balances		0	0	0		1	0	-1		-2	0	2

Example 5: Succession step 1

		X_1				X_2				Y		
		x_{11}	x_{12}	x_{13}		x_{21}	x_{22}	x_{23}		y_1	y_2	y_3
X_1	x_{11}	0	0	0		0	0	0		2	1	-3
	x_{12}	0	0	0		0	0	0		0	0	0
	x_{13}	0	0	0		0	0	0		0	0	0
X_2	x_{21}	0	0	0		0	0	0		-2	0	2
	x_{22}	0	0	0		0	0	0		0	0	0
	x_{23}	0	0	0		0	0	0		0	0	0
Y	y_1	1	2	-3		0	0	0		0	0	0
	y_2	0	0	0		1	0	-1		0	0	0
	y_3	0	0	0		0	0	0		0	0	0
Balances		0	0	0		0	0	0		-2	0	2

Example 6: Succession step 2

Succession analysis often leads to stable, internally consistent scenarios but may also lead to cycles. Example 7 adds cross-impacts for y_3 on X_1 and y_3 on Y, which generates a cycle starting from the scenario $\{x_{12}, x_{21}, y_3\}$ and proceeding through $\{\underline{x_{11}}, x_{21}, y_3\}$, $\{x_{11}, x_{21}, \underline{y_1}\}$, $\{\underline{x_{12}}, x_{21}, \underline{y_2}\}$, and back to $\{\underline{x_{12}}, x_{21}, \underline{y_3}\}$.

		X_1				X_2				Y		
		x_{11}	x_{12}	x_{13}		x_{21}	x_{22}	x_{23}		y_1	y_2	y_3
X_1	x_{11}	0	0	0		0	0	0		2	1	-3
	x_{12}	0	0	0		0	0	0		0	0	0
	x_{13}	0	0	0		0	0	0		0	0	0
X_2	x_{21}	0	0	0		0	0	0		-2	0	2
	x_{22}	0	0	0		0	0	0		0	0	0
	x_{23}	0	0	0		0	0	0		0	0	0
Y	y_1	1	2	-3		0	0	0		0	0	0
	y_2	0	0	0		1	0	-1		0	0	0
	y_3	**1**	**0**	**-1**		0	0	0		**1**	**-1**	**0**
Balances		1	0	-1		0	0	0		-1	-1	2

Example 7: Start of cycle

Descriptors often have tied variants. How ties are broken affects how a succession unfolds. Example 8 adds a cross-impact for y_3 on x_{12}, which creates a tie between x_{11} and x_{12} in the scenario $\{x_{12}, x_{21}, y_3\}$. How this tie is broken determines whether the succession ends at $\{x_{12}, x_{21}, y_3\}$ or whether it enters the cycle from Example 7. If the tie is broken in favour of x_{11}, the succession enters the cycle. When it returns to $\{x_{12}, x_{21}, y_3\}$, the tie must be broken again.

		X_1				X_2				Y		
		x_{11}	x_{12}	x_{13}		x_{21}	x_{22}	x_{23}		y_1	y_2	y_3
X_1	x_{11}	0	0	0		0	0	0		2	1	-3
	x_{12}	0	0	0		0	0	0		0	0	0
	x_{13}	0	0	0		0	0	0		0	0	0
X_2	x_{21}	0	0	0		0	0	0		-2	0	2
	x_{22}	0	0	0		0	0	0		0	0	0
	x_{23}	0	0	0		0	0	0		0	0	0
Y	y_1	1	2	-3		0	0	0		0	0	0
	y_2	0	0	0		1	0	-1		0	0	0
	y_3	1	**1**	-1		0	0	0		1	-1	0
Balances		1	1	-1		0	0	0		-1	-1	2

Example 8: Tied balances

1.6 Relative strength of cross-impacts

Within judgment groups, cross-impacts indicate the relative plausibility of the variants of a target descriptor. Within impact columns, cross-impacts have a different meaning, indicating the **relative strength** of the impacts of source variants on a target variant. In example 8, the judgment group defined by x_{11} and Y contains three cross-impacts: 2, 1, and -3. Within the judgment group, these cross-impacts mean that y_1 is most plausible, y_2 less so, and y_3 implausible. The impact column of y_1 contains three cross-impacts: 2, -2, and 1. Within the impact column, these cross-impacts mean that x_{11} has the strongest positive impact on y_1, x_{21} has an equally strong but negative impact on y_1, and y_3 has a positive but half as strong impact on y_1.

In summary, the cross-impacts within an impact column give the **relative strength** of the source variants and comprise a **strength distribution** for the impact column.

1.7 Rules of thumb for assigning cross-impacts

First, assign cross-impacts within judgment groups, focusing on the relative plausibility of the target variants given the source variant. Next, scan up and down the judgment groups within a target descriptor, iteratively adjusting the cross-impacts to achieve the desired relative strengths. Be careful to preserve the plausibility distributions within the judgment groups. An easy way to do this is to add, subtract, multiply, or divide all cross-impacts within a judgment group by the same value. For example, to weaken the relative strength of x_{21} on Y, the cross-impacts within the judgment group could be multiplied by 0.5. This transformation preserves the relative plausibility within the judgment group while adjusting the strength of x_{21} on Y relative to the other source variants.

1.8 Exception to zero-sum rule in case of equally (im)plausible variants

Sometimes two or more variants of a target descriptor Y are judged to be equally (im)plausible with respect to some variant of source descriptor X. In this case, the cross-impacts within the corresponding judgment group <u>must not</u> sum to zero across all variants individually. While two or more variants can be equally (im)plausible, only one of them can occur, and so only one of them needs to be compensated. This exception is necessary for preserving relative plausibility within judgment groups and allowing for relative strength within impact columns. Failure to make this exception results in overly strong compensating cross-impacts that can distort the relative strength of the target variants.

Examples 9 and 10 present the incorrect and correct specifications of negative compensating cross-impacts when two or more variants are equally plausible. The same principle applies symmetrically to positive compensating cross-impacts when two or more variants are equally implausible.

In Example 9, the negative compensating cross-impact on y_3 offsets the positive impacts on both y_1 and y_2 individually. This results in duplicated compensation and an overly strong negative constraint on y_3, causing y_2 to emerge as the unique variant of Y with the highest balance.

In Example 10, y_3 is correctly compared against only one of y_1 or y_2, since they are equally plausible and only one compensating offset is required. As a result, y_2 and y_3 are tied for the highest balance, opening up two succession paths and necessitating a tie-breaker rule. As these examples show, the correct specification of compensating cross-impacts when variants are equally (im)plausible directly affects the shape of the succession.

			Y		
			y_1	y_2	y_3
X_1	x_{11}		2	2	-4
	x_{12}		0	0	0
	x_{13}		0	0	0
X_2	x_{21}		-2	-1	3
	x_{22}		0	0	0
	x_{23}		0	0	0
Balances			0	1	-1

Example 9: Tied cross-impacts (incorrect specification)

			Y		
			y_1	y_2	y_3
X_1	x_{11}		2	2	-2
	x_{12}		0	0	0
	x_{13}		0	0	0
X_2	x_{21}		-2	-1	3
	x_{22}		0	0	0
	x_{23}		0	0	0
Balances			0	1	1

Example 10: Tied cross-impacts (correct specification)

1.9 Perturbation analysis

The set of all scenarios in a CIB model comprises a state space. Internally consistent scenarios and cycles are the attractors of the state space. Through succession analysis, basins of attraction can be identified, transforming the state space into a stability landscape. Perturbation analysis extends succession analysis, tracing how attractors evolve when a number their variants is replaced. The replaced variants are the **perturbations**, and the scenarios with the replacements are the **neighbours** of the attractor. The degree of perturbation to an attractor is measured by Hamming distance, which is the number of variants by which the attractor and the neighbour differ. For example, in Example 6, the scenario $\{x_{12}, x_{21}, y_3\}$ is internally consistent and therefore a point attractor. This attractor has six neighbours of degree 1:

- $\{\underline{x_{11}}, x_{21}, y_3\}$, x_{12} is replaced by x_{11}
- $\{\underline{x_{13}}, x_{21}, y_3\}$, x_{12} is replaced by x_{13}
- $\{x_{12}, \underline{x_{22}}, y_3\}$, x_{21} is replaced by x_{22}
- $\{x_{12}, \underline{x_{23}}, y_3\}$, x_{21} is replaced by x_{23}
- $\{x_{12}, x_{21}, \underline{y_1}\}$, y_3 is replaced by y_1
- $\{x_{12}, x_{21}, \underline{y_2}\}$, y_3 is replaced by y_2

Perturbations that result in a new attractor are **leverage points**, and the subsequence of scenarios from the neighbour to the attractor is a **pathway**. As mentioned above, the outcome of succession analysis on a neighbour depends on how ties are broken. Thus, multiple pathways and outcomes can be possible for a particular set of perturbations.

Part 2: Prophesy User Guide

2.0 Conventions

- Scenarios are reported as comma-separated strings (e.g., 1,2,4), where the numbers are the variant indices of the descriptors. The variants are reported in the same order as the descriptors are presented in the Settings form and on the Impact Matrix sheet.
- Successions are reported as chevron-separated strings of scenarios (e.g., 1,2,4 > 2,2,4 > 2,1,4 > 2,1,2).
- Succession trials are separated by semicolons (e.g., 1,2,4 > 2,2,4 > 2,1,4 > 2,1,2; 1,2,4 > 2,1,4 > 2,1,4 > 2,1,4).

2.1 Installing Prophesy

Prophesy is currently built as a cross-platform add-in for Excel, compatible with MacOS and Windows. To install:

- Download Prophesy from Zenodo at https://doi.org/10.5281/zenodo.7473654. You need the .xlam file.
- Open Excel.
- Go to the Developer tab.
    - If the Developer tab is not visible, go to File > Options > Customize Ribbon and check Developer, then click OK.
- Click Excel Add-ins.
- Click Browse.
- Find the Prophesy .xlam file, select it, and click OK.
- Click OK in the Add-ins window.
- Prophesy is now installed and a Prophesy tab appears in the ribbon.

To install a new version, repeat the steps above with the new file and deactivate the old version.

2.2 Creating a project

- Go to the Prophesy ribbon.
- Click *New Project*.
- A *Create New Project* pop-up appears. Click Yes.

- A *Save As* window appears.
- Navigate to the folder where you want your project to live.
- Enter a name for the project and click *Save*.
- A workbook with the name of your project is created, containing an empty "Impact Matrix" sheet.

2.3 Settings

2.3.1 Descriptors page

2.3.1.1 Adding and modifying descriptors and variants

- On the Prophesy ribbon, click *Settings*.
- The *Settings* window appears.
- On the *Descriptors* page, click *Add Descriptor*.
- An *Add Descriptor* window appears.
- Enter a name for the new descriptor and click *OK*.
- By default, new descriptors are created with two variants, named Variant1 and Variant2.
- To add additional variants, select the descriptor from the list of descriptors and click *Add Variant*.
- Descriptors and variants can be renamed, deleted, and reordered using the buttons on the *Descriptors* page.
- Reordering the descriptors or variants on the *Descriptors* page also reorders how the descriptors, variants, and their cross-impacts are presented on the *Impact Matrix* sheet.
- Clicking *Save and close* will create or update a matrix and the *Scenario Navigator* on the *Impact Matrix* sheet.

2.3.1.2 Types of descriptors

There are three types of descriptors in Prophesy: nominal, ordinal, and auxiliary. A descriptor's type affects how the descriptor is corrected during succession analysis.

2.3.1.2.1 Nominal descriptors

The descriptor can move directly from the current variant to the variant with the highest balance.

2.3.1.2.2 Ordinal descriptors

The descriptor can move only one variant per succession step in the direction of the variant with the highest balance.

Example

Let:

- Y = an ordinal descriptor with variants $\{y_1,y_2,y_3\}$.

If y_1 is the current variant of Y and y_3 is the variant of Y with the highest balance, then Y can only move from y_1 to y_2 in the first succession step. If y_3 remains the variant with the highest balance at the next succession step, then Y can move from y_2 to y_3.

2.3.1.2.3 Auxiliary descriptors

Auxiliary descriptors allow to represent impacts that depend on the variants of two or more nominal or ordinal descriptors. The variants of auxiliary descriptors are combinations of the variants of the descriptors on which they are based.

Example

Let:

- Y = a target descriptor with variants $\{y_1,y_2\}$
- X_1 = a source descriptor with variants $\{x_{11},x_{12}\}$
- X_2 = a source descriptor with variants $\{x_{21}, x_{22}\}$
- A = an auxiliary descriptor composed from X_1 and X_2, such that the variants of A are $\{x_{11}\&x_{21}, x_{11}\&x_{22}, x_{12}\&x_{21}, x_{12}\&x_{22}\}$

Suppose that y_1 is promoted if and only if $X_1 = x_{11}$ and $X_2 = x_{22}$. Then when:

- When $A = x_{11}\&x_{22}$

there is some cross-impact on y_1.

Auxiliary descriptors are updated and express their impacts in the same succession step as changes in the variants of the descriptors on which they are based.

2.3.1.2.4 Creating an auxiliary descriptor

- Create a new descriptor.

- Change its type to *Auxiliary*.
- An *Add Auxiliary Descriptor* window appears.
- Add two or more descriptors.
- Click *Save*.
- To modify an existing auxiliary descriptor, change its type to nominal or ordinal and then back to auxiliary, which will reopen the *Add Auxiliary Descriptor* window.
- The variants of auxiliary descriptors are not directly editable, as they are based on the selected descriptors.

2.3.1.3 Chance Random Variant

Each descriptor has a *Chance of random variant*, ranging from 0 to 100.

When a descriptor's *Chance of random variant* is greater than zero, at each succession step, there is a chance that the variant of the descriptor will be assigned at random.

This feature allows to represent inherent randomness or uncertainty about the behaviour of one or more descriptors.

2.3.1.4 RNG seed

At project creation, a seed for the random number generator is automatically specified and reported for reproducibility.

The user may edit the *RNG seed*.

The *RNG seed* is used to determine random events (e.g., breaking ties during succession analysis, creating model versions).

2.3.2 Scenarios page

2.3.2.1 Type of initial scenarios, solver, and number of initial scenarios

There are two types of initial scenario: procedurally generated and custom.

2.3.2.1.1 Procedurally generated

When the *Full* solver is selected, all possible scenarios are created and used for scenario analysis. The number of initial scenarios is therefore fixed and cannot be edited.

When the *Discrete Maximin* solver is selected, the specified number of scenarios is generated using a discrete maximin design. The aim is to produce a set of scenarios that is both balanced and diverse.

For each non-auxiliary descriptor, its variants are distributed as evenly as possible across the requested number of scenarios and then randomly permuted. This produces balanced marginal distributions for each descriptor.

The algorithm generates many candidate designs and selects the design that maximizes the minimum Hamming distance between scenarios (that is, the number of descriptors that differ between two scenarios). If several designs achieve the same minimum distance, the design with the largest average Hamming distance is selected. Maximizing the minimum Hamming distance ensures that scenarios are as different from one another as possible.

Auxiliary descriptors are assigned their default variant and are not used in the sampling procedure.

This approach produces a scenario set that is well distributed across the scenario space while respecting the requested number of scenarios.

2.3.2.1.2 Custom

When the type of initial scenarios is set to *Custom*, an *Open* window appears. Select the file containing the custom scenarios. The name of the file will be shown in the textbox adjacent to the dropdown menu.

The first row of the file must contain headers (descriptor names) and the subsequent rows contain the custom scenarios. Each column represents one descriptor. The variant indices are used rather than the variant names.

The solver is restricted to *Full* and the number of initial scenarios is not editable.

Descriptor 1	Descriptor 2	Descriptor 3	Descriptor ...
1	2	1	...
2	2	1	...
1	1	1	...
...	...	...	...

Correct formatting of custom scenarios file

2.3.3 Successions page

2.3.3.1 Tiebreaker rules

Two or more variants of a descriptor can be tied for the highest balance. There are three options for breaking ties: conservative, random, and legacy.

Conservative: When the current variant of a descriptor is tied with another variant, retain the current variant. Otherwise, use the random tiebreaker.

Random: When two or more variants of a descriptor are tied, select one of them at random.

Legacy: When two or more variants of a descriptor are tied, select the one with the lowest variant index.

2.3.3.2 Steps per succession trial

Specifies the number of steps per succession trial (greater than or equal to one).

2.3.3.3 Succession trials per initial scenario

Specifies the number of times to conduct a succession analysis per initial scenario (greater than or equal to one). When any feature that introduces randomness is used (conservative/random tiebreaker, chance random variant), multiple succession trials per initial scenario are recommended, as the results of the succession analysis can vary.

The requested number of succession trials is conducted for each initial scenario across each model version.

2.3.3.4 Minimum cycle transitions

Specifies the minimum number of transitions that can delimit a cycle. The minimum cycle transitions can take values between 2 and the maximum cycle transitions.

Example

Let

A > B > A > C > A > B > A > C > B > A

be a succession.

When minimum cycle transitions is 2, the cycle **A > B > A** is detected and occurs twice:

- [**A > B > A**] > C > A > B > A > C > B > A
- A > B > A > C > [**A > B > A**] > C > B > A

2.3.3.5 Maximum cycle transitions

Specifies the maximum number of transitions that can delimit a cycle. Maximum cycle transitions can take values between minimum cycle transitions and the total number of transitions in the succession trial, which is the number of steps minus one.

Example

Let

A > B > A > C > A > B > A > C > B > A

be a succession.

When minimum cycle transitions is 3 and maximum cycle transitions is 4, the short cycle **A > B > A** is not detected, but three longer cycles are detected, each occurring once:

- A > B > A > C > A > [**B > A > C > B**] > A
- A > B > A > C > A > B > [**A > C > B > A**]
- [**A > B > A > C > A**] > B > A > C > B > A

If maximum cycle transitions is decreased to 3, the last cycle is not detected.

2.3.3.6 Cycle noise tolerance

Succession analysis can produce point attractors (modal successors) and cyclical attractors (repeating sequences of successors). When any feature that introduces randomness is used (conservative/random tiebreaker, chance random variant), cyclical attractors can become noisy. Cycle noise tolerance allows for the detection of noisy cyclical attractors, where noise refers to the number of additional scenarios in the cycle that deviate from the clear pattern. Cycle noise tolerance can take values between 0 and (steps per succession trial - minimum cycle transitions − 1).

Example

Let

A > B > A > C > A > B > A > C > B > A

be a succession.

When minimum cycle transitions and maximum cycle transitions are 2 and cycle noise tolerance is 0, a clear cycle with 2 transitions, **A > B > A**, is detected and occurs twice:

- [**A > B > A**] > C > A > B > A > C > B > A
- A > B > A > C > [**A > B > A**] > C > B > A

When cycle noise tolerance is increased to 1, a third instance of **A > B > A** is detected:

- A > B > A > C > A > B > [**A > <u>C</u> > B > A**], where C is the noisy element.

2.3.4 Perturbations page

2.3.4.1 Source of scenarios

There are two options: last scenario analysis and custom.

2.3.4.1.1 Last scenario analysis

The default option; the file containing the most recently conducted scenario analysis within the project.

2.3.4.1.2 Custom

Select a file containing a list of scenarios to perturb (such as another scenario analysis conducted within the project).

The first row of the file must contain headers, and the subsequent rows contain the scenarios. The first column contains the scenarios as comma-separated lists of descriptor variants indices. The second and third columns may contain trial convergence and model convergence statistics (optional).

Modal Attractor	Trial Convergence	Model Convergence
1,2	0.5667	1
1,1	0.1167	0.5

Correct format of custom source of scenarios file

2.3.4.2 Degree of perturbation

Number of descriptor variants replaced simultaneously. Determines the number of neighbours per initial scenario. Can take integer values between 1 and the total number of descriptors in the model.

Let

- k = Degree of perturbation (number of descriptor variants replaced simultaneously)
- v = number of descriptors in the model
- L_i = number of variants of descriptor i
- N = number of neighbours per initial scenario
- S = subset of k descriptors selected from v descriptors

Then:

$$N = \sum_{\substack{S \subseteq \{1,\ldots,v\},\\ |S|=k}} \prod_{i \in S} (L_i - 1)$$

Example

Suppose that:

- k = 2 (two descriptor variants replaced simultaneously)
- v = 5
- $L_1 = L_2 = L_3 = L_4 = 3$
- $L_5 = 2$

Pairs among descriptors 1-4:

There are:

(4 choose 2) = 6

such pairs.

Each descriptor in these pairs has 3 – 1 = 2 alternative variants.

Each pair therefore generates:

2 x 2 = 4

neighbours.

Total from these pairs:

6 x 4 = 24

Pairs involving descriptor 5:

There are four such pairs:

(1,5), (2,5), (3,5), and (4,5)

Descriptor 5 has only 2 – 1 = 1 alternative variant.

Each such pair generates:

2 x 1 = 2

neighbours.

Total from these pairs:

4 x 2 = 8

Total neighbours

N = 24 + 8 = 32

Example neighbours

Suppose we have the following initial scenario:

{1,1,1,1,1}

For the descriptor pair (1,2), the neighbours are:

- {2,2,1,1,1}
- {2,1,2,1,1}
- {2,1,1,2,1}
- {2,1,1,1,2}

For the descriptor pair (1,5), the neighbours are:

- {2,1,1,1,2}
- {3,1,1,1,2}

And similarly for all other descriptor pairs.

2.3.4.3 Trials cutoff

The percentage of all succession trials across all model versions (global trial convergence) for which a scenario must be an attractor to be included in the perturbation analysis. Can take values between 0 and 100.

Example

Let A be an attractor with global trial convergence of 80%. When trials cutoff ≥ 70, A is included in the perturbation analysis.

2.3.4.4 Models cutoff

The percentage of model versions (global model convergence) for which a scenario must be an attractor in at least one succession trial to be included in the perturbation analysis. Can take values between 0 and 100.

Example

Let A be an attractor with global model convergence of 80%. When models cutoff ≥ 70, A is included the perturbation analysis.

2.3.5 Models page

2.3.5.1 Type of models and number of models

There are two types of models: procedurally generated and custom.

2.3.5.1.1 Procedurally generated

The specified number of model versions is generated from the base model on the *Impact Matrix* sheet using a discrete maximin design over judgment sections.

A judgment section corresponds to a pair of descriptors (source descriptor → target descriptor). All cross-impacts between those descriptors are treated together when generating model versions.

For each judgment section, Prophesy assigns one of several possible treatments:

- **Excluded:** The judgment section is not included in the model because it is excluded according to its confidence value in the *Confidence Matrix*.
- **Deleted:** The judgment section is included according to its confidence value but is subsequently removed according to *Proportion Delete*.
- **Included unchanged:** The judgment section is included and its cross-impacts are unchanged.
- **Sign flipped:** The judgment section is included and the signs of its cross-impacts are reversed.

- **Multiplied:** The judgment section is included and its cross-impacts are multiplied by the specified *Multiplier* value.
- **Sign flipped and multiplied:** The judgment section is included, its cross-impacts are multiplied by the specified *Multiplier*, and their signs are reversed.

The proportions of these treatments are controlled by the *Confidence Matrix* and the parameters *Proportion Delete*, *Proportion Flip Sign*, and *Proportion Multiply*. The confidence value determines the fraction of model versions in which a judgment section is included, while the other parameters determine how included sections are modified.

For each judgment section, the treatments are distributed as evenly as possible across the generated model versions according to the specified proportions. The treatments are then randomly permuted across model versions.

Prophesy generates many candidate model designs and selects the design that maximizes the minimum Hamming distance between model versions (that is, the number of judgment sections treated differently between two models). If several designs achieve the same minimum distance, the design with the largest average Hamming distance is selected. Maximizing the minimum Hamming distance ensures that model versions are as different from one another as possible.

This produces a diverse set of model versions while approximately preserving the requested proportions of deletions, sign flips, and multiplications.

All stochastic decisions are deterministic given the project's RNG seed, ensuring that results are reproducible.

2.3.5.1.2 Confidence matrix

When enabled, a *Confidence Matrix* sheet is added with default values of 1 for all judgment sections with non-zero cross-impacts in the *Impact Matrix* sheet.

The confidence value of a judgment section (in the range [0,1]) represents the fraction of generated model versions in which that judgment section is included.

When procedurally generating model versions, the treatments assigned to each judgment section are distributed so that approximately the specified proportion of models includes the section.

2.3.5.1.4 Custom

Select a file containing custom model versions. The model versions must have the same dimensions (total number of variants) as the base model on the *Impact Matrix* sheet. The first row and the first column must be empty. After the first model, each model version must be separated from the previous one by two empty rows. When the loading custom model versions, the number of models is fixed and cannot be edited.

2.3.5.2 Proportion Delete

Let PD be the proportion of judgment sections that are deleted.

- If PD = 0, no deletions occur.
- If PD > 0, approximately PD of included judgment sections are deleted in the generated models.

2.3.5.3 Proportion Flip Sign

Let PF be the proportion judgment sections whose signs are flipped.

- If PF = 0, no sign flips occur.
- If PF > 0, approximately PF of included and non-deleted judgment sections have their signs flipped in the generated models.

2.3.5.4 Proportion Multiply and Multiplier

Let M be a fixed multiplier (Double; default 1) and let PM be the proportion of judgment sections that are multiplied by M.

- If M = 1 or PM = 0, multiplication has no effect.
- Otherwise, approximately PM of included and non-deleted judgment sections are multiplied by M in the generated models.

2.4 Impact Matrix and Scenario Navigator

After at least one descriptor has been added, an impact matrix and the *Scenario Navigator* appear on the *Impact Matrix* sheet. The impact matrix is where the user inputs cross-impacts. The *Scenario Navigator* allows the user to manually explore scenarios. It consists of the following fields:

- **Current**: Located in the top-left of the *Impact Matrix* sheet, the user inputs a scenario in this field by manually selecting descriptor variants by their indices (e.g., 1,1,1).
- **Successor**: Located immediately below *Current*, this field shows the successor of the current scenario once *Load Scenario* has been clicked. When *Single Succession* is clicked, the successor replaces the current scenario and its successor is presented.
- **Balances**: Located immediately below the impact matrix, this field shows the balances of the current scenario once it has been loaded. The balances are updated each time *Load Scenario* or *Single Succession* is clicked.
- **Random Variant**: Located immediately below *Balances*, this field shows which variants, if any, were selected at random. The indicator is "RE".
- **Tiebreak**: Located immediately below *Random variant*, this field shows which variants, if any, were selected by a tiebreaker, and which tiebreaker was used. "CON" indicates the *Conservative* tiebreaker, *"RND"* indicates the *Random* indicator, and "LEG" indicates the *Legacy* tiebreaker.
- **Succession signature**: Located beneath the impact matrix, this field shows the initial scenario, its successors, and the count of random tiebreaks at each step in a succession trial. This field is updated every time *Load Scenario*, *Single Succession*, or *Full Succession* is clicked.

2.5 Scenario analysis

Scenario analysis computes succession trials of the specified length for initial scenarios over model versions and identifies:

- The attractor for each succession trial
- The modal attractor for each initial scenario across all its succession trials across all model versions
- Descriptive statistics and diagnostics

2.5.1 Setting up a scenario analysis

- Prepare your model:
    - Descriptors, including type, variants, and chance of random variant
    - Cross-impacts
- Configure the settings. All settings are relevant except the settings on the Perturbations page.

- Click *Scenario Analysis*.

2.5.2 Results sheets

Scenario analysis generates eight sheets: Global Attractor Summary, Attractor Matrix, Succession Trials, Random Tiebreaks Matrix, Trial Stability Matrix, Trial Coverage Matrix, Cycle Repeats Matrix, and Project Details.

2.5.2.1 Global Attractor Summary

Reports all attractors found during the analysis. The first column contains the attractors. The next seven columns contain descriptive statistics: Trial Convergence, Model Convergence, Trial Convergence within Relevant Models, Model IDs, Average Stability, Average Trial Coverage, and Average Cycle Repeats. The statistics are calculated for each attractor across all succession trials across all model versions.

2.5.2.2 Attractor Matrix

Reports the attractors for the succession trials. The first column contains the initial scenarios. The subsequent columns contain the results per model version. Each cell contains the string of attractors associated with the succession trials of the initial scenario and the model version. Point attractors are modal scenarios (strings of descriptor indices separated by commas). Cyclical attractors are strings of scenarios separated by chevrons. The attractors are separated by semi-colons.

Example of a point attractor: 1,2,3

Example of a cyclical attractor: 1,2,3 > 1,1,1 > 1,2,3

2.5.2.3 Succession Trials

Reports the full details of the succession trials. The first column contains the initial scenarios. The second column contains the modal attractor for the initial scenarios across their succession trials across all model versions. The third to fifth columns contain descriptive statistics: Trial Convergence, Model Convergence, Model IDs, Average Stability, Average Trial Coverage, and Average Cycle Repeats. The statistics are calculated *for each initial scenario* across its succession trials across all model versions. The subsequent columns contain the results per model version. Each cell contains the string of succession

trials associated with the initial scenario and the model version. The scenarios within succession trials are separated by chevrons. The succession trials are separated by semi-colons.

2.5.2.4 Random Tiebreaks Matrix

Reports the count of descriptors with variants tied for highest balance for all succession trials*. The first column contains the initial scenarios. The subsequent columns contain the values per model version. Each cell contains the strings of tie counts for each step in each succession trial associated with the initial scenario and the model version; the tie count for the initial scenario is prepended to each string**. Steps are separated by commas, and trials are separated by semi-colons. These values are used in calculating *Trial Stability*.

* Only applies when using the *Conservative* or *Random* tiebreaker.

** The initial scenario is included because ties in the starting state contribute to instability within the succession trial (see below).

2.5.2.5 Trial Stability Matrix

Reports the *Trial Stability* for all succession trials.

Let:

- N = number of descriptors
- SST = number of steps per succession trial
- STI = number of succession trials per initial scenario
- $R_{i,s}$ = number of random tiebreaks required at step s of trial i
- V_0 = number of descriptors with ties in the initial scenario
- TS be the Trial Stability for a succession trial

Then:

$$TS = 1 - \frac{(V_0 * STI) + \sum_{i=1}^{STI} \sum_{s=1}^{SST-1} R_{i,s}}{N * SST * STI}$$

The initial scenario is included because ties in the starting state contribute to instability within the succession trial. The last step of each succession trial is excluded because once the system reaches that step, no further branching can occur within that trial, so it cannot contribute to instability.

The first column contains the initial scenarios. The subsequent columns contain the values per model version. Each cell contains the string of *Trial Stability* values for the succession trials associated with the initial scenario and the model version. Individual trials are separated by semi-colons. These values are used in calculating *Average Stability*.

Example

A succession analysis is conducted with ten steps per succession trial over ten succession trials per initial scenario for a model with seven descriptors. For a particular initial scenario, one random tiebreak is required and across the succession trials no random tiebreaks are required.

$$TS = 1 - \frac{(1 * 10) + 0}{7 * 10 * 10} \approx 0.9857$$

2.5.2.6 Trial Coverage Matrix

Reports the *Trial Coverage* for all succession trials. *Trial Coverage* is the number of steps in a succession trial occupied by the attractor of the trial. The first column contains the initial scenarios. The subsequent columns contain the values per model version. Each cell contains the string of trial coverages associated with the initial scenario and the model version. Trials are separated by semi-colons. These values are used in calculating *Average Trial Coverage*.

2.5.2.7 Cycle Occurrences Matrix

Reports the number of times a cycle occurs for all succession trials. The first column contains the initials scenarios. The subsequent columns contain the values per model version. Each cell contains the number of cycle occurrences for each succession trial associated with the initial scenario and the model version. Individual trials are separated by semi-colons. These values are used in calculating *Average Cycle Occurrence*s.

2.5.2.8 Project Details

Reports the settings used for the analysis.

2.6 Perturbation analysis

Perturbation analysis generates neighbours of initial scenarios, computes succession trials of the specified length for neighbours over model versions, and identifies:

- The attractor for each succession trial
- The modal attractor for each neighbour across all its succession trials across all model versions
- The perturbations (variant replacements) that cause an initial scenario (via a neighbour, which is the initial scenario modified by the perturbations) to evolve into an attractor
- Descriptive statistics and diagnostics

Cycles are split into their scenarios and included in perturbation analysis so that the effects of perturbations at different steps in a cycle can be observed.

The basis for a perturbation analysis is a scenario analysis or custom list of initial scenarios, which need not be attractors (e.g., perturbing a succession to observe what happens at each step).

2.6.1 Setting up a perturbation analysis

- Select your initial scenarios:
    - Last scenario analysis, or
    - Custom
- Configure the settings. All settings are relevant. The most important setting is the Degree of perturbation, which affects how many neighbours are generated per initial scenario.
- Click *Perturbation Analysis*.

2.6.2 Results sheets

Perturbation analysis generates the same eight sheets as scenario analysis, with two additional columns for the neighbours and the perturbations. A ninth sheet is also generated: Transformation Matrix.

2.6.2.1 Transformation Matrix

Reports the leverage points (effective perturbations) that cause an initial scenario to evolve into an attractor. The left column of the matrix contains the initial scenarios. The top row of the matrix contains the attractors. The remaining cells contain the string of leverage points associated with the initial scenario and the attractor. Leverage points are reported by their descriptor and variant indices. Combinations are joined by &. Leverage points are followed by a hyphen and the indices of the model versions in which they occur. Multiple leverage points are separated by semi-colons.

Example

	1,2,3,1,1
1,1,1,1,1	1(2)&2(1)-1,10; 1(2)&3(3)-6

This transformation matrix is read as follows:

- Simultaneously changing Descriptor 1 to its second variant *and* Descriptor 2 to its first variant causes the initial scenario {1,1,1,1,1} to evolve into the attractor {1,2,3,1,1} in models 1 and 10.
- Simultaneously changing Descriptor 1 to its second variant *and* Descriptor 3 to its third variant causes the initial scenario {1,1,1,1,1} to evolve into the attractor {1,2,3,1,1} in models 6.

The transformation matrix is heat mapped by similarity: similar pairs of initial scenario and attractor have a light shading, while dissimilar pairs have a darker shading.

2.7 Pathway analysis

Pathway analysis generates a list of pathways, descriptive statistics, and scores (optionally) for a pair of source scenario and target scenario. The basis for a pathway analysis is a perturbation analysis.

2.7.1 Setting up a pathway analysis

- Click *Pathway Analysis*.
- The most recent perturbation analysis is automatically loaded, if any. Click *Select file* to load the results of a different perturbation analysis.

- Select a source and a target scenario. The options are taken from the *Transformation Matrix* of the selected perturbation analysis.
- Pathways may be scored by the variants that occur (e.g., for desirability). To create a scoring profile, click *Add*. An *Add or Edit Scoring Profile* appears. Select a descriptor and a variant, specify a score (integer and fractional values are accepted), and click *Add*. When ready, click *Save and close*. You will be prompted to give the scoring profile a name. To use the scoring profile, select it from the list of scoring profiles in the *Pathway Analysis* form.
- When ready, click *Search*. If it is the first search using the selected perturbation analysis file, you will be prompted to create a name for the pathway analysis.
- When a source scenario is selected, only target scenarios that can be accessed by a leverage point in the associated transformation matrix are shown. Likewise, when a target scenario is selected, only source scenarios from which the target scenario can be accessed by a leverage point in the associated transformation matrix are shown. To see all source and target scenarios, click *Clear search*.

2.7.2 Results sheet

Each search creates a new sheet, which reports the source scenario, the target scenario, the scoring profile used (if any) and its ingredients, and the list of pathways with summary statistics and scores (if requested).

2.8 Identification of attractors

For each succession trial, Prophesy identifies the attractor that best characterizes the dynamics of the succession. Two possible attractor types are considered:

- Cyclical attractors (repeated cycles of scenarios)
- Point attractors (modal scenarios)

The attractor whose coverage of the succession trial is greatest is selected as the attractor for the succession trial. If two or more attractors are tied for greatest coverage of the succession trial, a tie is reported.

2.8.1 Cycle detection

Let a succession trial be a sequence of scenarios

$$S = (s_1, s_2, \dots, s_T)$$

where T is the number of steps.

A cycle candidate is a subsequence

$$(s_i, s_{i+1}, \dots, s_j)$$

such that

$$s_i = s_j$$

and the subsequence contains at least two distinct scenarios.

The cycle length is defined as the number of transitions in the loop:

$$L = j - i$$

Only candidates satisfying

$$L_{min} \leq L \leq L_{max}$$

are considered.

Cycle identity

Cycle candidates that differ only by rotation are treated as the same cycle.

Example

A > B > A

B > A > B

Both represent the same alternating cycle between A and B. Cycles are converted to a canonical representation using the lexicographically smallest rotation of the cycle sequence.

Detecting occurrences

For each cycle candidate, Prophesy searches the entire succession trial for occurrences of the cycle.

An occurrence is detected when the sequence of transitions of the candidate cycle can be matched within the succession trial.

Occurrences may:

- appear anywhere in the trial
- overlap (shared indices are allowed)

Cycle Noise tolerance

The user may specify a cycle noise tolerance parameter.

Cycle noise tolerance allows additional scenarios to appear inside a cycle occurrence while still matching the cycle pattern.

Example

Cycle candidate:

A > B > A

With cycle noise tolerance = 1, the following sequence counts as an occurrence:

A > C > B > A

because the additional scenario C is treated as noise.

Cycle noise tolerance therefore expands the number of succession steps that may be consumed when matching an occurrence, but it does not change the cycle length.

Cycle repetition

Let

$$O$$

be the number of occurrences of a cycle candidate in the succession trial.

A cycle candidate is promoted to a cycle only if

$$O \geq 2$$

i.e. the cycle repeats at least once.

Cycle coverage

The coverage of a cycle is defined as

$$C = L \times O$$

where

- L= number of transitions in the cycle
- O= number of detected occurrences

This measure reflects how strongly the succession dynamics follow the cycle pattern.

Determining the tentative cycle winner

If multiple cycles are detected, the cycle with the highest coverage $C$is the tentative cycle winner.

If two or more different cycles have equal coverage, the cycle result for the trial is:

Tied Cycles within Trial

2.8.2 Modal scenario identification

Let

$$f(x)$$

be the number of occurrences of scenario $x$in the succession trial.

The modal scenario is

$$m = \arg\max_{x} f(x)$$

If the maximum frequency is unique, the modal scenario is the unique scenario with the highest frequency.

If multiple scenarios share the maximum frequency, the modal result is:

Tied Modals within Trial

2.8.3 Modal–cycle comparison

Let

- C = coverage of the tentative cycle winner
- M = frequency of the modal scenario

The final trial signature is determined as follows:

Condition	**Result**
Cycle comparison result = Tied Cycles within Trial and (C > M)	Tied Cycles within Trial
Cycle comparison result = Tied Cycles within Trial and (C = M)	Tied Modal and Cycle within Trial
Cycle comparison result = Tied Cycles within Trial and (C < M)	Modal result
(C > M)	Cycle
(C = M)	Tied Modal and Cycle within Trial
(C < M) and modal scenario unique	Modal scenario
(C < M) and modal scenarios tied	Tied Modals within Trial

2.9 Summary statistics and diagnostics

Each succession trial has an attractor, which is the modal scenario of the succession trial (point attractor), the modal cycle of the succession trial (cyclical attractor), or a tie. Each initial scenario has a modal attractor, which is the attractor that occurred most frequently across its succession trials across all model versions.

Prophesy generates a set of summary statistics for the modal attractors (*Succession Trials* sheet), all attractors (*Global Modal Summary* sheet), and for pathways.

2.9.1 Modal attractor statistics

These statistics appear on the *Succession Trials* sheet for scenario analyses and perturbation analyses. They are calculated for modal attractors of initial scenarios, not for all attractors.

- **Trial Convergence:** The proportion of succession trials across all model versions for which the modal attractor of the initial scenario is the attractor of the succession trial. This statistic is analogous to the "attractor weight" for deterministic CIB.
- **Model Convergence:** The proportion of model versions in which the modal attractor of the initial scenario is the attractor of at least one succession trial.
    - **Model IDs:** The IDs of the model versions in which the modal attractor of the initial scenario is the attractor of at least one succession trial.
- **Trial Convergence within Relevant Models:** The proportion of succession trials for which the modal attractor of the initial scenario is the attractor of the succession trial, considering only those models in which the modal attractor is the attractor of at least one succession trial.
- **Average Stability:** The average stability across all model versions, where stability is a measure of the determinism of the initial scenario and its succession trials, considering only those succession trials in which the modal attractor of the initial scenario is the attractor of the succession trial.
- **Average Trial Coverage:** The average trial coverage across all model versions, where trial coverage is the number of steps in a succession trial occupied by the modal attractor of the initial scenario, considering only those succession trials in which the modal attractor of the initial scenario is the attractor of the succession trial. Each scenario in a cycle contributes to trial convergence.
- **Average Cycle Occurrences:** The average number of times the (cyclical) modal attractor of the initial scenario occurred within the succession trials for which it was the attractor across all model versions.

2.9.2 All attractor statistics

These statistics appear on the *Global Attractor Summary* sheet for scenario analyses and perturbation analyses. They are calculated for all attractors, not for modal attractors of initial scenarios.

- **Trial Convergence:** The proportion of succession trials across all model versions for which the attractor is the attractor of the succession trial. This statistic is analogous to the "attractor weight" for deterministic CIB.

- **Model Convergence:** The proportion of model versions in which the attractor is the attractor of at least one succession trial.
    - **Model IDs:** The IDs of the model versions in which the attractor is the attractor of at least one succession trial.
- **Trial Convergence within Relevant Models:** The proportion of succession trials for which the attractor is the attractor of the succession trial, considering only those models in which the attractor is the attractor of at least one succession trial.
- **Average Stability:** The average stability across all model versions, where stability is a measure of the determinism of the initial scenario and its succession trials, considering only those succession trials in which the attractor is the attractor of the succession trial.
- **Average Trial Coverage:** The average trial coverage across all model versions, where trial coverage is the number of steps in a succession trial occupied by the attractor, considering only those succession trials in which the attractor is the attractor of the succession trial. Each scenario in a cycle contributes to trial convergence.
- **Average Cycle Occurrences:** The average number of times a cyclical attractor occurred within the succession trials for which it was the attractor across all model versions.

2.9.3 Pathway statistics

These statistics appear on the *Pathways Summary* sheet for pathway analyses. They are calculated for the pathways, where a pathway is a subsequence of scenarios starting after the initial scenario and ending at the attractor (in the case of cycles, ending at any scenario within the cycle).

- **Count:** The number of succession trials across all model versions containing the pathway.
- **Share (Perturbation):** The proportion of succession trials across all model versions containing the pathway, considering only those succession trials that contain the associated perturbation. When stochastic settings are used (conservative or random tiebreaker or chance of random variant), the same perturbation can generate multiple pathways, and so this proportion can be less than 100.
- **Share (All Pathways):** The proportion of all succession trials across all model versions containing the pathway.
- **Model convergence:** The proportion of model versions in which at least one succession trial contains the pathway.

    - **Model IDs:** The IDs of the model versions in which the pathway occurs in at least one succession trial.
- **Average stability:** The average stability of the initial scenario together with the pathway across all model versions, considering only those succession trials that contain the pathway.
- **Min Score:** The score of the lowest-scored scenario in the pathway.
- **Max Score:** The score of the highest-scored scenario in the pathway.
- **Initial Score:** The score of the initial scenario before the start of the pathway.
- **Final Score:** The score of the attractor at the end of the pathway.
- **Cumulative Score:** The sum of the scores along the pathway.
    - **Score Trace:** A string of scores for the scenarios along the pathway.

2.9.4 Diagnostics

When the Balanced Maximin solver is used, the following diagnostics appear on the *Global Attractor Summary* sheet:

- **Design iterations:** The number of random shuffles of the descriptor columns used to generate candidate scenario designs. Each descriptor has a column with rows equal to the number of specified scenarios. Within each column, the descriptor's variants are assigned approximately equally (maximum difference of 1 between counts). Each shuffle produces a different combination of variants across the rows. Increasing the number of iterations increases the chances of finding a design with larger minimum and average Hamming distance between scenarios.
- **Number of distinct scenarios:** The number of distinct scenarios in the selected design. When the number of sampled scenarios is large relative to the total number of possible scenarios, duplicate scenarios may occur, resulting in a minimum Hamming distance of 0 and poorer coverage of the scenario space.
- **Minimum Hamming distance:** The smallest Hamming distance among all pairs of scenarios in the selected design. The Hamming distance between two scenarios is the number of descriptors for which the scenarios have different variants. Designs with larger minimum Hamming distance are preferred because they ensure that even the most similar pair of scenarios differ in several descriptors, increasing the likelihood that scenarios fall into different basins of attraction and aiding attractor discovery.
- **Average Hamming distance:** The average Hamming distance across all pairs of scenarios in the selected design. It is used to break ties among designs with equal

minimum Hamming distance. Designs with larger average Hamming distance are preferred because they produce more diverse scenarios overall, improving coverage of the scenario space and aiding scenario discovery.

- **Pairwise variant coverage:** The average fraction of possible variant combinations observed for each pair of descriptors across the selected scenarios. For a pair of descriptors with L_1 and L_2 variants, there are $L_1 \times L_2$ possible variant combinations. Pairwise variant coverage measures how many of these combinations appear in the scenario set. The reported value is the average coverage across all pairs of descriptors. Designs with higher pairwise variant coverage explore more combinations of descriptor variants, improving coverage of the scenario space and increasing the chances of identifying distinct basins of attraction.

When multiple model versions are generated, the following diagnostics appear on the *Global Attractor Summary* sheet:

- **Design iterations:** The number of random shuffles of the judgment-section columns used to generate candidate model designs. Each judgment section has a column with rows equal to the number of generated model versions (excluding the base model). Within each column, the possible treatments of the judgment section (for example exclusion, deletion, sign flip, or multiplication) are assigned approximately according to their specified proportions. Each shuffle produces a different combination of treatments across the model versions. Increasing the number of iterations increases the chances of finding a design with larger minimum and average Hamming distances between model versions.
- **Number of distinct model versions:** The number of distinct model versions in the selected design. When the number of generated model versions is large relative to the number of possible treatment combinations, duplicate model versions may occur, resulting in a minimum Hamming distance of 0 and poorer coverage of the space of possible model variations.
- **Minimum Hamming distance:** The smallest Hamming distance among all pairs of model versions in the selected design. The Hamming distance between two model versions is the number of judgment sections for which the treatments differ. Designs with larger minimum Hamming distance are preferred because they ensure that even the most similar pair of model versions differs in several judgment sections, increasing the diversity of model structures explored and aiding sensitivity analysis and the assessment of attractor robustness across model versions.

- **Average Hamming distance:** The average Hamming distance across all pairs of model versions in the selected design. It is used to break ties among designs with equal minimum Hamming distance. Designs with larger average Hamming distance are preferred because they produce more diverse model versions overall, improving coverage of the space of possible model variations and aiding sensitivity analysis.

Part 3: Tutorials

These tutorials explore the implications of a theory about how competition at multiple levels (among individuals, among firms, and among states) affect economic growth, environmental degradation, and identity, and how these factors interact with one another. The theory is elaborated in Kearney (2024).

In brief, the theory posits that competition among states for resources, competition among firms for profit, and competition among individuals for income and goods are mutually reinforcing. Each of them drives economic growth, environmental degradation, and social polarization (in-group favouritism and out-group derogation). In turn, social polarization drives competition at all levels, and economic growth and environmental degradation drive competition among states. The theory explores the potential for ecological identity – cognitively, affectively, and experientially identifying with the rest of nature – to circumvent these dynamics and create potential for more cooperative social norms and institutions.

3.1 Scenario analysis

Create a model with the following **nominal** descriptors[2]:

Descriptor	**Variants**
Domestic economic growth	High, Medium, Low
Interstate competition for resources	High, Medium, Low
Interfirm competition for profits	High, Medium, Low
Interpersonal competition for income and goods	High, Medium, Low
Environmental degradation	High, Medium, Low
Social identity	Extensive, Moderate, Limited
Ecological identity	Extensive, Moderate, Limited

[2] For ideas that inspired this example model, see Kearney, N. (2024). "Human Dignity and Ecological Identity." In *Policy Sciences and the Human Dignity Gap*, edited by S.G. Clark, E.J. Andrews, and A.E. Lambert. Natural Resource Management and Policy, vol 58. Spinger, Cham. https://doi.org/10.1007/978-3-031-52501-8_16.

Add this pattern of cross-impacts

3	-1	-2
1.5	-0.5	-1
0	0	0

to the following judgment sections:

- Interstate competition → Domestic economic growth
- Interfirm competition → Domestic economic growth
- Interpersonal competition → Domestic economic growth
- Domestic economic growth → Interstate competition
- Interfirm competition → Interstate competition
- Environmental degradation → Interstate competition
- Interpersonal competition → Interfirm competition
- Interfirm competition → Interpersonal competition
- Interstate competition → Environmental degradation
- Interfirm competition → Environmental degradation
- Interpersonal competition → Environmental degradation
- Ecological identity → Social identity

Add this pattern of cross-impacts

-2	-1	3
-1	-0.5	1.5
3	-1	-2

to the following judgment sections:

- Social identity → Interstate competition
- Social identity → Interfirm competition
- Social identity → Interpersonal competition

Add this pattern of cross-impacts

-2	-1	3
-1	-0.5	1.5
0	0	0

to the following judgment sections:

- Ecological identity → Environmental degradation
- Interstate competition → Social identity

- Interfirm competition → Social identity
- Interpersonal competition → Social identity

Configure the following settings (leave the other settings to their default values):

Descriptors	**Scenarios**	**Successions**	**Models**
RNG seed = 1	Solver = Full	Tiebreaker = Conservative	Number of model versions = 1
		Steps per succession trial = 10	
		Succession trials per initial scenario = 10	
		Minimum cycle transitions = 2	
		Maximum cycle transitions = 3	
		Cycle noise tolerance = 0	

Analyze single scenarios

Go to the *Impact Matrix* sheet. In the *Current* row at the top-left, replace the zeroes with ones and click *Load Scenario*. Which descriptors are inconsistent? What happens if you click *Single Succession*?

Now enter the following scenario: 3,3,3,3,1,3,1. Click *Load Scenario*. Click *Single Succession* repeatedly until you reach an internally consistent scenario. Enter the scenario and load it again. Now click *Full Succession*. Look at *Succession Signature* under the matrix. The initial scenario and its ten successors are shown (steps per succession trial = 10).

Analyze all scenarios

There are 3^7 = 2'187 scenarios in this model. With ten steps per succession trial and ten succession trials per initial scenario, that translates to 2'187'000. Analyzing this volume of scenarios manually would be impractical. Click *Scenario Analysis*. Name the analysis, "SA_con". The progress of the analysis is shown in the status bar at the bottom of the workbook. The runtime will be about seven minutes. When the analysis finishes, examine the *Global Attractor Summary* sheet. Out of the 2'187 possible scenarios, how many are attractors for the model?

Experimenting with the settings

Change the *Tiebreaker* setting to *Random*. Go to the *Impact Matrix* sheet and enter 1,1,1,1,1,1,1 for *Current*. Click through the succession trial with *Single Succession*. Do you get the same results as when you used the *Conservative tiebreaker?*

Run another *Scenario Analysis* and name it "SA_rand". The runtime will be about ten minutes. When it finishes, examine the *Global Attractor Summary* sheet. What do you notice? Look at the quantity and the type of attractors. Why do you think you got cycles this time?

Find attractors that are in the *Global Attractor Summary* for both SA_con and SA_rand and compare their descriptive statistics. What do you notice? Look at their *Trial Convergence* and *Average Stability*.

Sensitivity analysis

The relative strength of the source variants is uniform. Are the results of the scenario analysis sensitive to small changes in the relative strengths? To find out, make the following changes to the settings:

Scenarios	**Successions**	**Models**
Solver = Balanced Maximin	Tiebreaker = Conservative	Number of model versions = 11
Number of initial scenarios = 219*		Proportion to Delete = 25
Design iterations = 0		Design iterations = 0

* (Click OK to ignore the warning about the sample size)

This will generate ten model versions, which will be tested along with the base model on the *Impact Matrix* sheet. The summary statistics will be calculated across all model versions. In the sample of generated model version, roughly 25% of the judgment sections with cross-impacts will be deleted.

Analyzing the full set of scenarios across 11 model versions would take some time, as Prophesy would need to analyze 2.4057 million scenarios. To make this part of the tutorial faster, a sample of 219 scenarios will be generated.

Because Prophesy uses a space-filling design to generate samples (Balanced Maximin), there is a good chance that the model versions and scenarios will be will distributed within their respective spaces. For model versions, good coverage of the sampling space provides better sensitivity analysis, capturing a wider variety of possible modifications to the base model. For scenario analysis, good coverage of the scenario space increases the chances of finding the model's attractors.

Click *Scenario Analysis* and name it "SA_con_multi". The runtime will be about fourteen minutes. When the analysis finishes, examine the *Global Attractor Summary*. What do you notice? Find the attractor following attractor in SA_con: 1,1,3,3,3,1,1. Was it found to be an attractor in all sixteen model versions in SA_con_multi? Look at its *Model Convergence*. In which model versions was it found to be an attractor? How confident can we be about whether this scenario is an attractor for the system that we are modelling?

3.2 Perturbation analysis

Open Settings and go to Perturbations. Change *Source of scenarios* to *Custom* and select the "SA_con" file from Tutorial 1.

Configure the following settings:

Descriptors	**Scenarios**	**Successions**	**Perturbations**	**Models**
RNG seed = 1	Solver = Full	Tiebreaker = Conservative	Degree of perturbation = 1	Number of model versions = 1
		Steps per succession trial = 10	Trials cutoff = 0	

		Succession trials per initial scenario = 10	Models cutoff = 0	
		Minimum cycle transitions = 2		
		Maximum cycle transitions = 3		
		Cycle noise tolerance = 0		

Click *Perturbation Analysis*. Name the analysis, "PA_1". The progress of the analysis is shown in the status bar at the bottom of the workbook. The runtime will be about two minutes. When the analysis finishes, examine the *Transformations Matrix* sheet. Find the intersection of the initial scenario {1,1,1,1,1,3,1} and the attractor {1,1,1,3,3,1,1}. Are there any leverage points? What does this mean?

Change *Degree of perturbation* to 2 and run another perturbation analysis, naming the analysis "PA_2". The runtime will be about five minutes. When the analysis finishes, find the same intersection as above. Are there any leverage points? What is the first scenario that has leverage points leading to {1,1,3,3,3,1,1}?

Create a new workbook. At A1, type the header, "Initial Scenario". At A2, enter "1,1,1,1,1,3,1", and at A3, enter "1,1,3,3,1,1,1" (without quotation marks). Save the file in the same folder as your project and name the file "pa3_scenarios". Your sheet should look like this:

Initial Scenario
1,1,1,1,1,3,1
1,1,3,3,1,1,1

In *Settings* go to *Perturbations* and change *Degree of perturbation* to 3, *Source of scenarios* to *Custom* (if it is currently set to *Custom*, switch it to *Procedurally generated* and then back to *Custom* to trigger the Open file dialog), and select pa3_scenarios. Run perturbation analysis, naming the analysis "PA_3". Click *OK* to ignore the warning about *Trial Convergence* and *Model Convergence*. The runtime will be about two minutes (if we had used all the initial

scenarios from SA_con, it would have taken about fifteen minutes). What do you notice? Are there any direct or indirect pathways from the initial scenario {1,1,1,1,1,3,1} to the attractor {1,1,1,3,3,1,1}?

3.3 Pathway analysis

Click *Pathway Analysis*, click *Select File*, and select PA_3. Under *Source Scenario*, select "1,1,1,1,1,3,1", and under *Target Scenario*, select "1,1,3,3,1,1,1".

Click *Add* to add a scoring profile. Add the following scores:

- Environmental degradation (High): -3
- Environmental degradation (Medium): -1
- Environmental degradation (Low): +2
- Social identity (Extensive): +2
- Social identity (Limited): -1
- Ecological identity (Extensive): +1
- Ecological identity (Limited): -0.5

Click *Save and close*. Name the profile "ENV_SI_EI. Select the profile.

Click *Search*. The analysis will take seconds. How many pathways are there from the source scenario to the target scenario?

Click *Pathway Analysis* again and change *Source Scenario* to 1,1,3,3,1,1,1 and *Target Scenario* to 1,1,3,3,3,1,1. Click *Search*.

Based on the scores, what are the best pathways:

- From {1,1,1,1,1,3,1} to {1,1,3,3,1,1,1}
- From {1,1,3,3,1,1,1} to {1,1,3,3,3,1,1}

Which perturbations do the best-scored pathways use? Are the best-scored pathways the most stable pathways?

A helpful way of visualizing a succession is in a tableau. In the results of the pathway analysis for {1,1,1,1,1,3,1} to {1,1,3,3,1,1,1}, find the following pathway:

1,3,1,3,1,1,1 > 1,1,3,3,1,1,1

In tableau format:

	Domestic economic growth	Interstate competition for resources	Interfirm competition for profits	Interpersonal competition for income and goods	Environmental degradation	Social identity	Ecological identity
Initial scenario	1	1	1	1	1	3	1
Neighbour	1	3	1	3	1	1	1
Attractor	1	1	3	3	1	1	1

Replacing the variant indices with their labels, we obtain a more easily interpretable tableau:

	Domestic economic growth	Interstate competition for resources	Interfirm competition for profits	Interpersonal competition for income and goods	Environmental degradation	Social identity	Ecological identity
Initial scenario	High	High	High	High	High	Limited	Extensive
Neighbour	High	Low	High	Low	High	Extensive	Extensive
Attractor	High	High	Low	Low	High	Extensive	Extensive

Shading the succession provides further clarity. Here we indicate the perturbations in black and subsequent changes in orange.

	Domestic economic growth	Interstate competition for resources	Interfirm competition for profits	Interpersonal competition for income and goods	Environmental degradation	Social identity	Ecological identity
Initial scenario	High	High	High	High	High	Limited	Extensive
Neighbour	High	Low	High	Low	High	Extensive	Extensive
Attractor	High	High	Low	Low	High	Extensive	Extensive

The following story emerges: three simultaneous perturbations are applied to the initial scenario (i.e., the neighbour): shifting interstate competition for resources from high to low, interpersonal competition for income and goods from high to low, and social identity from limited to extensive. In the subsequent step of the succession, interstate competition for resources reverses from low to high, while interfirm competition for profits shifts from high to low. The system stabilizes here at an attractor.

Next, we explore a pathway from {1,1,3,3,1,1,1} to {1,1,3,3,3,1,1}, such as:

3,3,3,3,1,2,1 > 3,1,3,3,3,1,1 > 1,1,3,3,3,1,1

We obtain the following tableau (converted from indices to labels and highlighted):

	Domestic economic growth	**Interstate competition for resources**	**Interfirm competition for profits**	**Interpersonal competition for income and goods**	**Environmental degradation**	**Social identity**	**Ecological identity**
Initial scenario	High	High	Low	Low	High	Extensive	Extensive
Neighbour	Low	Low	Low	Low	High	Moderate	Extensive
Intermediate successor	Low	High	Low	Low	Low	Extensive	Extensive
Attractor	High	High	Low	Low	Low	Extensive	Extensive

This tableau reveals the following story: three simultaneous perturbations are applied to the initial scenario (i.e., the neighbour): shifting interstate competition for resources from high to low, interfirm competition for profits from high to low, and social identity from extensive to moderate. In the subsequent step of the succession, interstate competition for resources reverses from low to high, while environmental degradation shifts from high to low and social identity reverses from moderate to extensive. In the final step of the succession, domestic economic growth reverses from low to high. The system stabilizes here at an attractor.

Taking these stories together we arrive at the following interpretation:

- Moving from the basin of {1,1,1,1,1,3,1} to {1,1,3,3,3,1,1} is a two-stage process.
- In the first stage, a set of three perturbations achieves two lasting changes:
    - Interfirm competition for profits shifts from high to low
    - Social identity shifts from limited to extensive.
    - Interstate competition for resources is resilient, reversing from low to high.
- In the second stage, a different set of three perturbations is applied to the attractor that was reached in the previous stage. All three changes are reversed, but they trigger a shift in environmental degradation from high to low.

Here we only looked at the results of PA_3, but there are also pathways from {1,1,3,3,1,1,1} to {1,1,3,3,3,1,1} (the second stage) that require only two perturbations, which is more efficient. For example, we can leave social identity in the extensive variant and only change domestic economic growth and interstate competition for resources.

Part 4: Development

4.1 Roadmap

Prophesy is under active development. New versions will be released regularly to fix bugs add functionality.

Near-term improvements include:

Perturbation Analysis

- Add a Capture statistic to the Transformation Matrix.
- Remove heat mapping of ties and add heat mapping for cycles.

General

- Align format of custom scenarios for Scenario *Analysis* and *Perturbation Analysis*.
- Continued refactoring and documenting of the code.
- Additional examples and teaching materials.

Long-term development plans include:

- Transitioning the core logic to an app-independent Python engine.
- Shifting the Excel add-in to a front-end only.
- Add a front-end for LibreOffice Calc to make Prophesy fully open access and for technological sovereignty (not dependent on Microsoft).
- Possible development of a standalone app.

4.2 Known issues

- On Windows, you might need to right click on the .xlam file, go to Properties, and under “Security”, click the “Unblock” checkbox to allow the add-in to run. This also requires fully closing and reopening Excel and may require deactivating and reactivating Prophesy (see below).
- The .xlam may not load properly from the Downloads folder. Place the .xlam file in another local folder.
- If the add-in stops working properly, you can try deactivating it in Excel Add-ins, closing Excel fully, reopening Excel, and reactivating Prophesy in Excel Add-ins. This usually restores functionality.

4.3 Release history

Beta versions of Prophesy for CIB have been released periodically since December 2022. Beta versions were indicated by the letter b after the version number. The first full release of the software took place on March 12, 2026.

4.4 Updates

To see the current version of Prophesy, click the *About* button in the ribbon. New versions are released on GitHub and archived on Zenodo with a DOI.

5. Acknowledgements

Prophesy has been a labour of love since 2021. During my PhD defence, Prof. Dr. Garry Peterson, my external examiner, suggested I develop my own CIB software for perturbation and pathway analysis. I took his suggestion and began working on Prophesy during my first postdoc position at the University of Bern, where my work was funded by Prof. Dr. Thomas Breu. My aim with Prophesy has been to build on the pioneering work of Prof. Dr. Wolfgang Weimer-Jehle, the founder of this brilliant method. My PhD supervisor, Prof. Dr. Vanessa Schwiezer, introduced me to CIB and deftly guided me through the development of my ideas about perturbation analysis and pathway analysis. Prof. Dr. Homer-Dixon, who co-supervised my PhD work, introduced me to complexity science, which enriched my engagement with CIB. Over the years, I have profited from conversations and collaborations with several wonderful colleagues, including Dr. Jude Kurniawan, Dr. Natalya Siddhantakar, Dr. Ajar Sharma, Dr. Anita Luzurko, Dr. Hannah Kosow, Dr. Wolfgang Hauser, and Dr. Mert Duygan. Special thanks to the students and colleagues of my *Systems Modelling for Sustainable Development* course at the University of Bern for their thought-provoking questions and inspiring engagements with CIB.

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